Introduction to Theoretical Computer Science

CS2490 - Session 18: Non-regular Languages and the Pumping Lemma

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Fall 2026

Septembre 28, 2026

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More ways to prove that a language is regular.

Today: how to prove that a language is not regular?

Pumping Lemma

Theorem

Let Σ be an alphabet. If L is a regular language over Σ, then there exists an integer $p \geq 1$ (the pumping length) such that, for every word $w \in L$ with $|w| \geq p$, there exists a factorization $w = xyz$ with $x, y, z \in \Sigma^*$ such that:

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Don't mess up the orders of the quantifiers!

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For every word $w \in L$ with $|w| \geq p$: consider the computation of $\mathcal{A}$ on input w: $q_0 \xrightarrow{w_1} q_1 \xrightarrow{w_2} q_2 \xrightarrow{w_3} \cdots \xrightarrow{w_n} q_n$.

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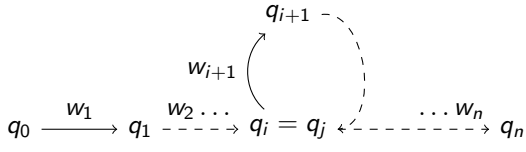
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So the run is really of the form:



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This concludes the proof of the Pumping Lemma.

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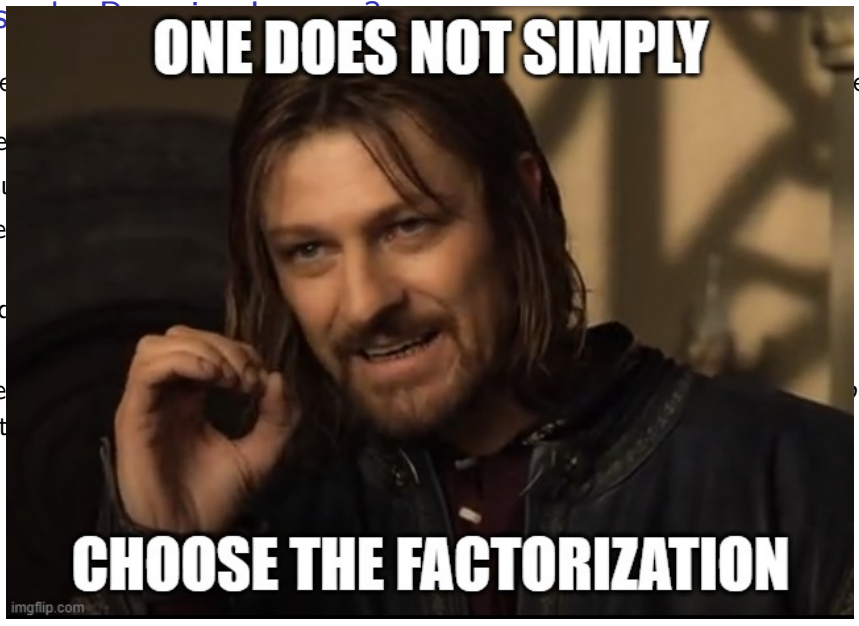
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- We found a contradiction, so L is not regular.

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More precisely, we design a winning strategy to beat the opponent's moves, whatever they may be.

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Question: Could we have chosen $w = a^p a^p$ instead?

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